Palm OS Programming Recipes

Recipe: Buttons

Most applications running on Palm OS systems use command buttons.

To add a button with Constructor:

1. drag and drop the button from the Catalog onto your Main form (the below code assumes the form's name is Main).
2. Set the Object Identifier. The below code refers to a button with an Object Identifier of "Test".
3. Set the label to be whatever you want. If need be adjust the width of the button such that the label is all visable.

There are 3 events generated that relate to a user interacting with a button on the screen.

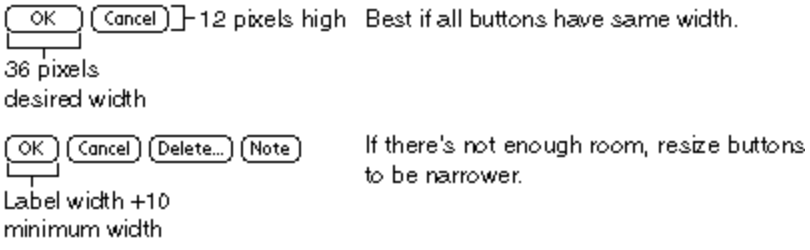
- a ctlEnterEvent is generated when the user puts the pen down within the bounds of the button
- a ctlExitEvent is generated after a ctlEnterEvent if the user pulls the pen up outside of the bounds of a button
- a ctlSelectEvent is generated after a ctlEnterEvent if the user pulls the pen up within the bounds of the button

Generally, you are only concerned with the third one, ctlSelectEvent. Add the following to your form's form handler to check for ctlSelectEvents.

```
if (event->eType == ctlSelectEvent) {
   switch (event->data.ctlSelect.controlID) {
      case MainTestButton:
         CreateNewObject();
         handled = true;
         break;

      case MainDetailsButton:
         ...
```

Buttons are left-justified at the bottom of a form. Each button should have the same width if there is space. If there isn't enough horizontal space or if one button's label is much longer than the others, then the buttons can have different widths. Try to size buttons so that they have 5 pixels of white space on either side of the button label. If the button must be smaller than that, ensure at least that there's an equal amount of white space on the left and right sides of the button label.

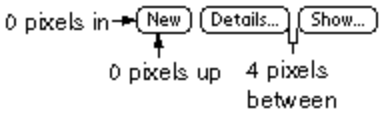


An "OK" button must have that label, not "Ok" or "ok". There should be an ellipse ("...") on a button that displays a dialog in order to perform its action, such as "Details..." It is better to use three periods than an actual ellipse character because that character does not exist in all character sets.

Proper placement of buttons is a subtle topic, but it can have a profound impact on the proper appearance of an application. You place buttons differently depending on whether the buttons appear on a standard (full-screen) form or on a modal dialog.

Placing Buttons on a Standard Form

On a form with no borders (that is, a full-screen form), the left-most button should begin at the left and bottom edges of the screen. Allow 4 pixels of white space between each button.

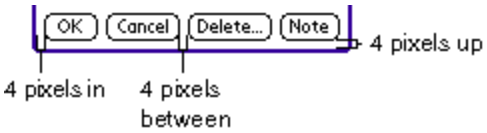


Borders are drawn outside of an object's coordinates, so you must allow space for the border. If you have a button 36 pixels wide and a left origin of 1, the button's left border is drawn at pixel 0 and the button's right border is drawn at pixel 37. The following table shows the proper button coordinates for a standard form. These coordinates might not make immediate sense; however, they are easily verified by looking at the screen at runtime. The preview of the screen provided by Constuctor or another visual layout tool isn't necessarily going to match the actual device precisely, so be sure to test with the Palm OS Emulator or a real device.

Dimension	Value
Left	1 (first button) For all other buttons, use the formula: *button(n-1) left + button(n-1) width + 6* to allow 4 pixels of white space plus 1 pixel for each button's border
Top	147
Width	36 if possible
Height	12

Placing Buttons on a Modal Dialog

On a modal dialog, you must have 4 pixels of white space between the form's border and the leftmost button's border to visually separate the two elements.



Dimension	Value
Left	5 (first button) 5 allows for 4 pixels white space + 1 pixel button border button border. For all other buttons, use the formula: *button(n-1) left + button(n-1) width* + 6 to allow 4 pixels of white space plus 1 pixel for each button's border
Top	Use the formula: *dialog height - button height* - 5
Width	36 if possible
Height	12